

DB Normalization - Exercices

Sistemas de Informação 1

Medication Form

Medication Form							
Patient number:		P1003		Department code:		Ort01	
Patient name:		John Smith		Department name:		Orthopedics	
Medication code	Medication name	Description	Dosage mg/ml	Administration method	Units per day	Start date	End date
10223	Morfina	analgesic	10	Oral	50	24-03-16	24-04-16
10351	Tetracycline	antibiotic	0.5	IV	10	24-03-16	17-04-16
10223	Morfina	analgesic	10	Oral	10	25-04-16	28-04-16

Medication Form

Patient number	Patient name	Dep. Code	Dep. name	Med. code	Med. name	Desc	Dosage	Admini. method	Units per day	Start date	End date
P1003	John Smith	Ort01	Orthopedics	10223	Morfina	analgesic	10	Oral	50	24-03-16	24-04-16
				10351	Tetracycline	antibiotic	0.5	IV	10	24-03-16	17-04-16
				10223	Morfina	analgesic	10	Oral	10	25-04-16	28-04-16

Medication Form – 1NF

- Set the primary key of the initial structure.
 - When we define the primary key, the structure can be considered in First Normal Form (1NF), assuming all attributes are atomic.
- Looking at the initial structure, the primary key corresponds to the attributes highlighted (underlined and in red):
 - Medication form (patient_number, medication_code, start_date, end_date, patient_first_name, patient_last_name, department_code, department_name, medication_name, description, dosage, administration_method, units_per_day).

Medication Form – 1NF

Patient number	First name	Last name	Dep. Code	Dep. name	Med. code	Med. name	Desc	Dosage	Admini. method	Units per day	Start date	End date
P1003	John	Smith	Ort01	Orthopedics	10223	Morfina	analgesic	10	Oral	50	24-03-16	24-04-16
P1003	John	Smith	Ort01	Orthopedics	10351	Tetracycline	antibiotic	0.5	IV	10	24-03-16	17-04-16
P1003	John	Smith	Ort01	Orthopedics	10223	Morfina	analgesic	10	Oral	10	25-04-16	28-04-16

Medication Form – 2NF

- The relation is in First Normal Form (1NF), and all non-key attributes must **functionally depend** on the entire primary key (full functional dependency).
- This requirement is only relevant **when the primary key is composite**.
 - In this case, it is composite.

Medication Form – 2NF

- When we analyse all non-key attributes with respect to the primary key, we identify some partial functional dependencies.
- They are:
 - patient_number → patient_first_name, patient_last_name, department_code, department_name
 - medication_code → medication_name, description, dosage, administration_method
 - (patient_number, medication_code, start_date, end_date) → units_per_day
 - This is the only attribute that depends on the entire primary key.

Medication Form – 2NF

- So, the final structure is:
 - Administration_Medication = (patient_number, medication_code, start_date, end_date, units_per_day);
 - Medicine = (medication_code, medication_name, description, dosage, administration_method);
 - Patient (patient_number, first_name, last_name, department_code, department name)

Medication Form – 2NF

Table Name: Patient (Patient number, Patient first name, Patient last Name, Department code, Department name)

patient_number	first_name	last_name	dep_code	dep_name
P1003	John	Smith	Ort01	Orthopedics

Table Name: Medicine (Medication code, Medication name, Description, Dosage, Administration method)

med_code	med_name	desc	dosage	admin_method
10223	Morfina	analgesic	10	Oral
10351	Tetracycline	antibiotic	0.5	IV

Table Name: Medication_Administration (Patient number, Medication code, Units per day, Start date, End date)

patient_number	med_code	units_per_day	start_date	end_date
P1003	10223	50	24-03-16	24-04-16
P1003	10351	10	24-03-16	17-04-16
P1003	10223	10	25-04-16	28-04-16

Medication Form – 3NF

- A relation is in Third Normal Form (3NF) if no non-key attribute **functionally depends** on another non-key attribute (i.e., there are no transitive dependencies).
- All relations are analysed, regardless of whether the primary key is simple (single attribute) or composite (multiple attributes).
- So:
 - The *Administration_Medication* table, because it has only one attribute that is not part of the key, is in 3NF;
 - In the *Medicine* table, all non-key attributes do not depend on each other, so it is in 3NF;
 - The Patient table is not 3NF, because **department_code** **determines** the **department_name**.

Medication Form – 3NF

Table Name: Patient (Patient number, Patient first name, Patient last Name)

patient_number	first_name	last_name
P1003	John	Smith

Table Name: Department (Department code, Department name)

dep_code	dep_name
Ort01	Orthopedics

Table Name: Medicine (Medication code, Medication name, Description, Dosage, Administration method)

med_code	med_name	desc	dosage	admin_method
10223	Morfina	analgesic	10	Oral
10351	Tetracycline	antibiotic	0.5	IV

Table Name: Medication form (Patient number, Department code, Medication code, Units per day, Start date, End date)

patient_number	dep_code	med_code	units_per_day	start_date	end_date
P1003	Ort01	10223	50	24-03-16	24-04-16
P1003	Ort01	10351	10	24-03-16	17-04-16
P1003	Ort01	10223	10	25-04-16	28-04-16